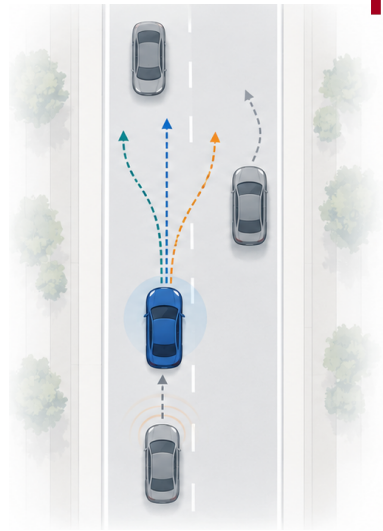
GhostCAV

From Frozen Driving Priors to Traffic Influence in Mixed Traffic

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October 5, 2026



02 | Background: from preferences to mechanisms

Mixed traffic: one vehicle's choices can change how surrounding drivers react.

Goal: use a frozen driving model to improve traffic while preserving ego performance and safety.

Early trajectory injection was unstable; the design moved to the prior's own candidates.

Midterm

SMART candidates → Reactive closed-loop outcomes → Learn preferences and reweight

Straight-road signal, close to random reweighting; no intersection advantage.

↓

Before expanding the learner, validate what the choices can actually change.

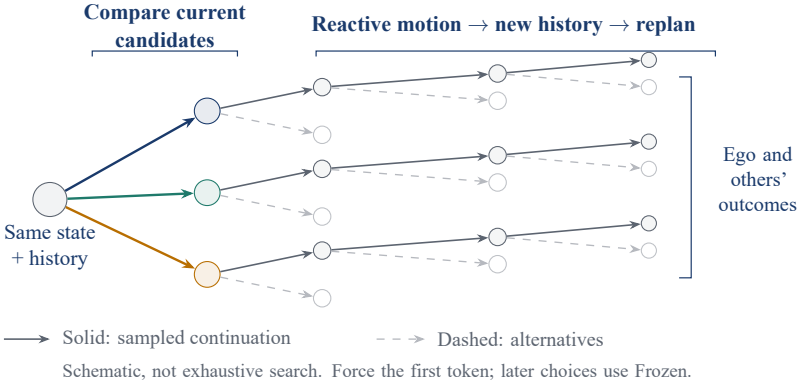
Post-midterm

Candidate diversity ✓ → Executable actions ✓ → Repeated ego control ✓ → Traffic influence ✓ → Reliable benefit ?

Focus: corridor following and lane changes. Intersection negotiation remains unvalidated.

03 | Method: evaluating choices in reactive traffic

Frozen SMART predicts the next 0.5 s motion from the map and updated history.



Goal: seek a choice that helps others while preserving ego performance and safety.

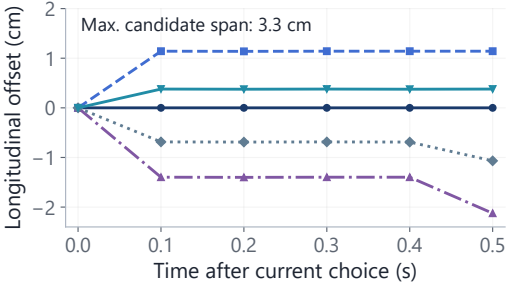
Each rollout requires traffic simulation and repeated model inference.

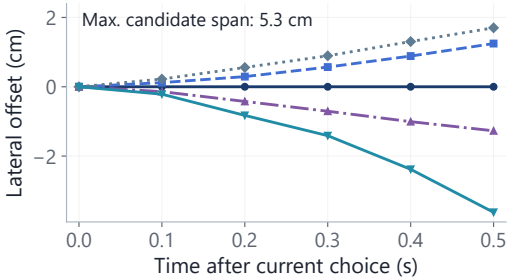
Opportunity check: 9 fixed states → 2 discovered pairs → 1 confirmed direction.
Both discoveries came from one road-and-traffic setup; learner expansion paused.

04 | Actions: does diversity survive execution?

Earlier, eight candidates executed identically; I traced and fixed a coordinate bug and an executor limit.



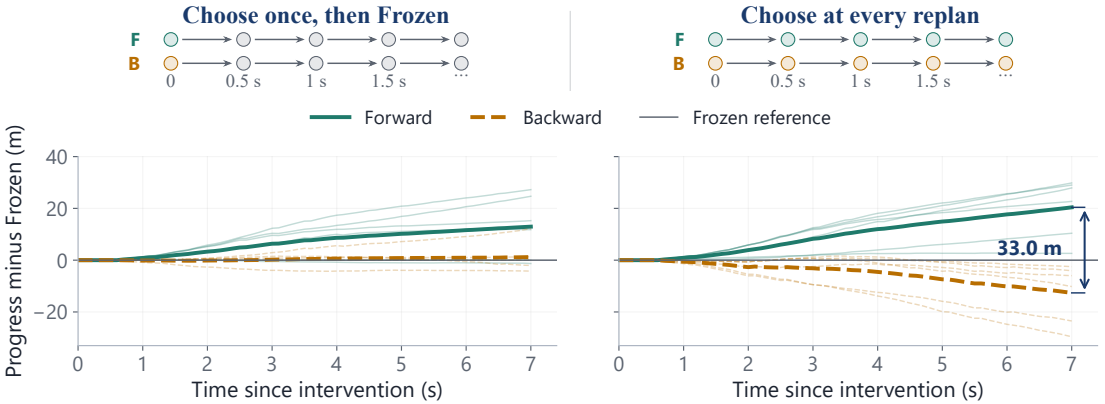




Later validated runs: five distinct Top-5 actions at each of three fixed starting states.
Offsets are relative to Rank 1 execution; short-action differences remain modest.

05 | Control: from one choice to repeated decisions

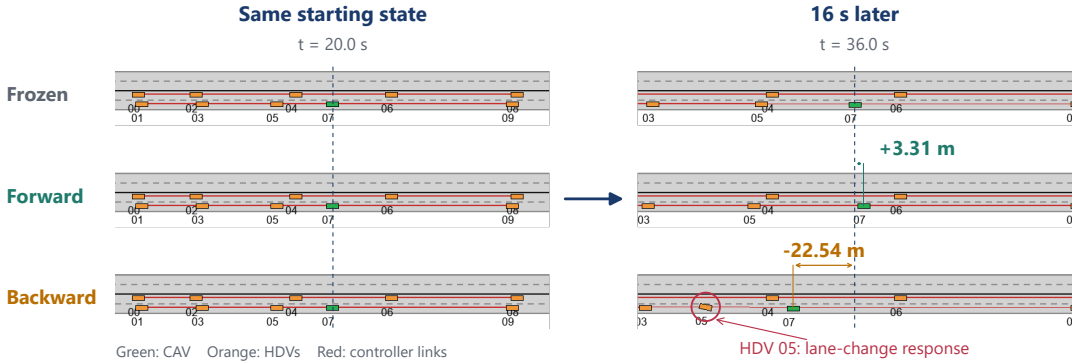
At each choice: Forward takes the largest native forward displacement in Top-5, Backward the smallest.



Repeated Forward stayed ahead of Frozen, Backward behind: 6/6 continuations in each of four setups.
Mean forward-minus-backward separation at 7 s: **12.7–33.0 m**.

06 | Transmission: same start, different traffic

At each 0.5 s replan, Forward / Backward choose the largest / smallest native forward motion in Top-5.



Control logs track the response: follower input → applied control → speed

HDV 05's lane-change decision was accepted at 34.3 s; the maneuver is visible at 36.0 s.

06 | Transmission: results across six cases

Two road geometries × three reused driver seeds. Response: changed applied control and speed.

Road	Driver seed	Response	Forward	Backward	Recorded event
Straight	82101	✓	+0.57	+4.03	None
Straight	82102	✓	−1.86*	+0.80	**Forward collision**
Straight	82103	✓	−0.87	+5.60	None
Curve	82101	✓	−4.16	+0.09	None
Curve	82102	✓	−4.56	−0.14	None
Curve	82103	✓	−2.56	+5.55	None

Forward / Backward: change in others' time loss (seconds) relative to Frozen. Negative means less loss.

*** Collision at 17.9 s: not acceptable traffic benefit.**

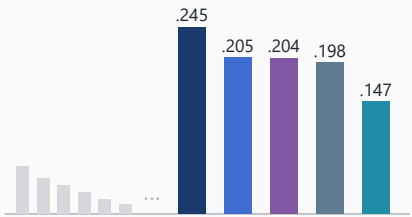
The starred comparison uses the common pre-event window of 17.8 s.

Traffic response occurred in all six cases; efficiency direction was mixed

Nine fixed other vehicles; 20 s except the starred window. None: no recorded collision/off-road. Controlled vehicle fixed before outcomes.

07 | Selection: reweight by predicted consequences

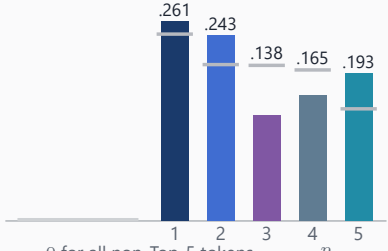
Frozen prior p



$$q_a \propto p_a \, e^{-\lambda \hat{L}_a}$$

toward lower predicted others' loss

Reweighted q



1 **Propose** Top-5 → **2** **Evaluate** each: force, then Frozen 3.5 s → **3** **Reweight** → **4** **Sample**, execute 0.5 s, replan

$$\min_q \; \mathbb{E}_q[\hat{L}] \quad \text{s.t.} \quad \mathrm{KL}(q \,\|\, p) \le \varepsilon, \;\; \mathbb{E}_q[\hat{J}] \ge \mathbb{E}_p[\hat{J}], \;\; \mathbb{E}_q[\hat{c}] \ge \mathbb{E}_p[\hat{c}]$$

$\hat{L}$: others' time loss $\hat{J}$: ego progress $\hat{c}$: minimum clearance

Stay close to the prior while shifting mass toward lower predicted others' loss.

07 | Selection: the loop runs, benefit does not follow

Results

Policy	Ego progress (m)	Others' loss (s)
Frozen	47.79	8.916
Ego-guided	49.34	8.910
Traffic-guided	**45.62**	**9.048**

One straight case, 7 s, matched randomness. Nine fixed other vehicles; lower loss is better.

First Traffic-guided decision

- Predicted losses differ by only **7 ms**, yet the full KL budget is spent.
- The draw changes; the follower responds, and the effect propagates in the adverse direction.

A good local choice need not make a good trajectory

Branches score one forced choice followed by Frozen; the policy keeps guiding. Local counterfactual preference did not translate into beneficial repeated behavior here.

Hard argmax on q — a different policy outside the KL budget — is not yet tested.

08 | Limitations and next steps

Current boundary	Next step
Reliable benefit remains open.	Calibrate predicted consequences and test repeated-policy benefit. Preserve the prior when evidence is weak.
Two road shapes, controlled car-following traffic.	Test held-out traffic and drivers, then broaden response models and interactions.
Full counterfactual branches are expensive.	Validate lightweight prediction for accuracy and decision cost.

Built: closed-loop counterfactual platform. **Established:** choice → ego control → traffic. **Open:** reliable benefit.

I will continue developing this research with my mentor, aiming for submissions to IEEE IV 2027 and IROS 2027.

Thank you

Questions welcome.

2026 MIT-NEU-UF Summer Camp Final Presentation